

COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

Computer Vision Laboratory

Lab Sheet – Experiment No. 2

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

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Course: B.Tech CSE

Academic Session: 2026–27

Labsheet. 2

Experiment Name

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Aim

To analyze and implement contrast enhancement techniques using histogram analysis, contrast stretching, Histogram Equalization, and CLAHE in Python and OpenCV for improving image quality and enhancing visual information in low-contrast images.

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Purpose

Image contrast is one of the most important factors affecting computer vision performance. Images captured under poor lighting, foggy weather, or weak camera sensors often have low contrast, making object detection and feature extraction difficult.

Histogram-based enhancement techniques improve the distribution of pixel intensity values, making important details more visible. These preprocessing methods are widely used before image segmentation, object recognition, OCR, and deep learning tasks.

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Learning Outcomes

After completing this experiment, students will be able to:

- Understand the concept of image contrast.
- Analyze pixel intensity using histograms.
- Perform contrast stretching.
- Apply Histogram Equalization.
- Implement CLAHE.

- Compare enhancement techniques.
- Interpret histogram changes after processing.
- Understand real-world applications.

Theory

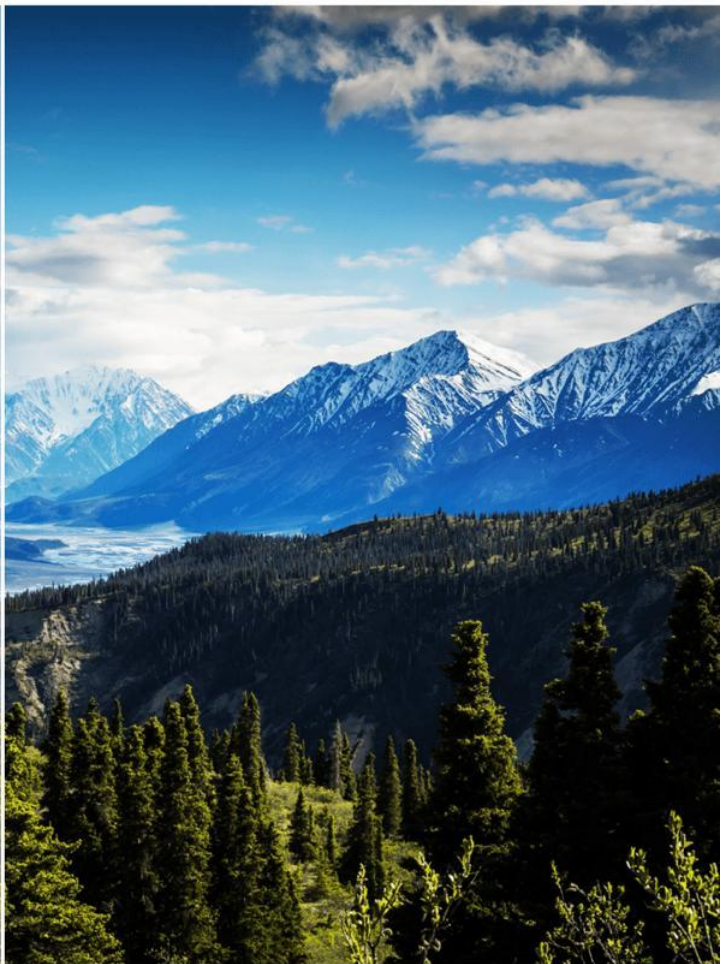
What is Image Contrast?

Image contrast is the difference between the darkest and brightest pixels of an image.

- High contrast → Clear details.
- Low contrast → Dull appearance.
- Good contrast improves visibility.



Low contrast



High contrast

Example

Low Contrast

Dull image

High Contrast

Sharp image

Hidden details

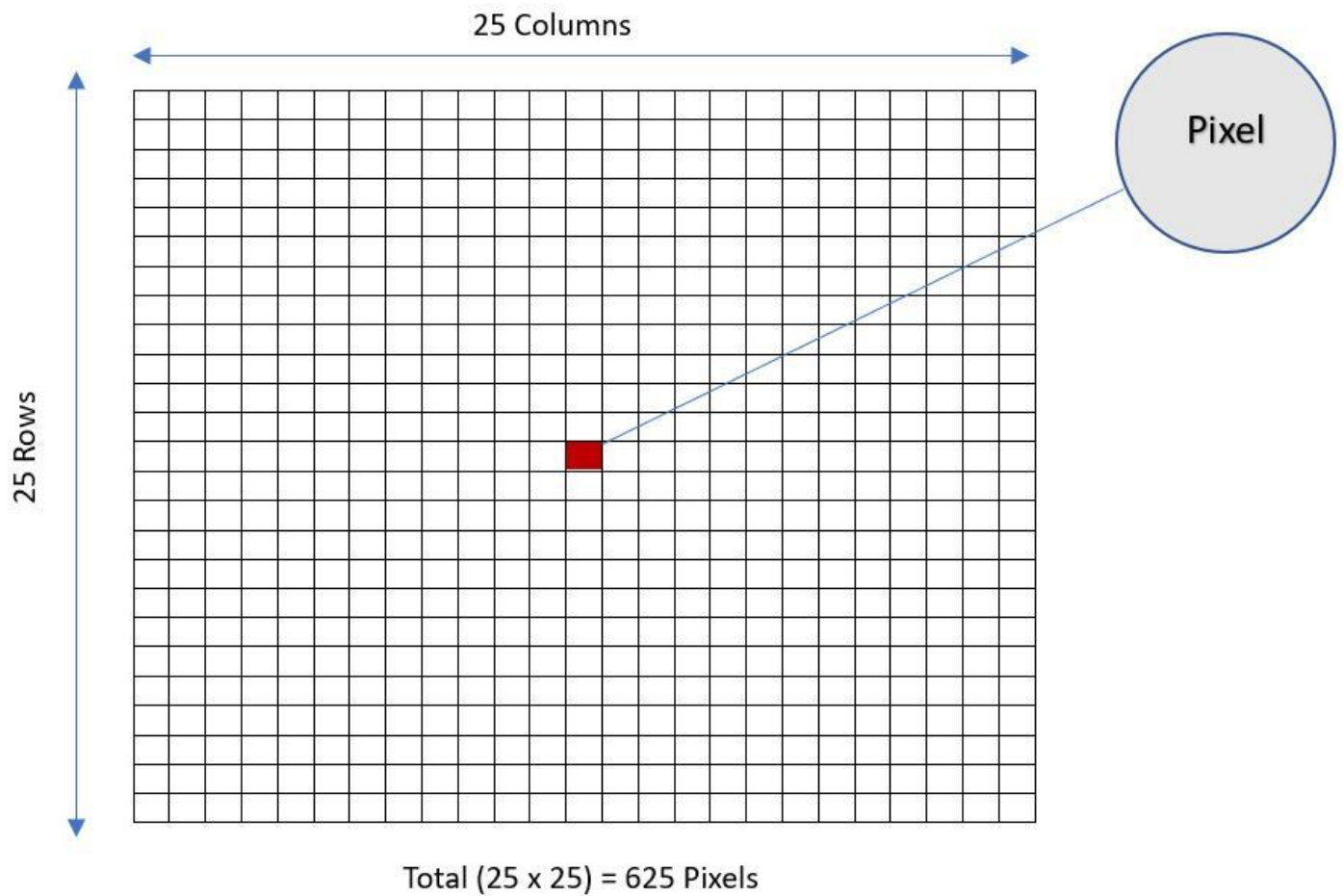
Visible details

Narrow intensity range Wide intensity range

Digital Image Representation

A digital image consists of thousands or millions of tiny squares called pixels.

Each pixel stores an intensity value.



Pixel Intensity

For grayscale images:

Pixel Value Meaning

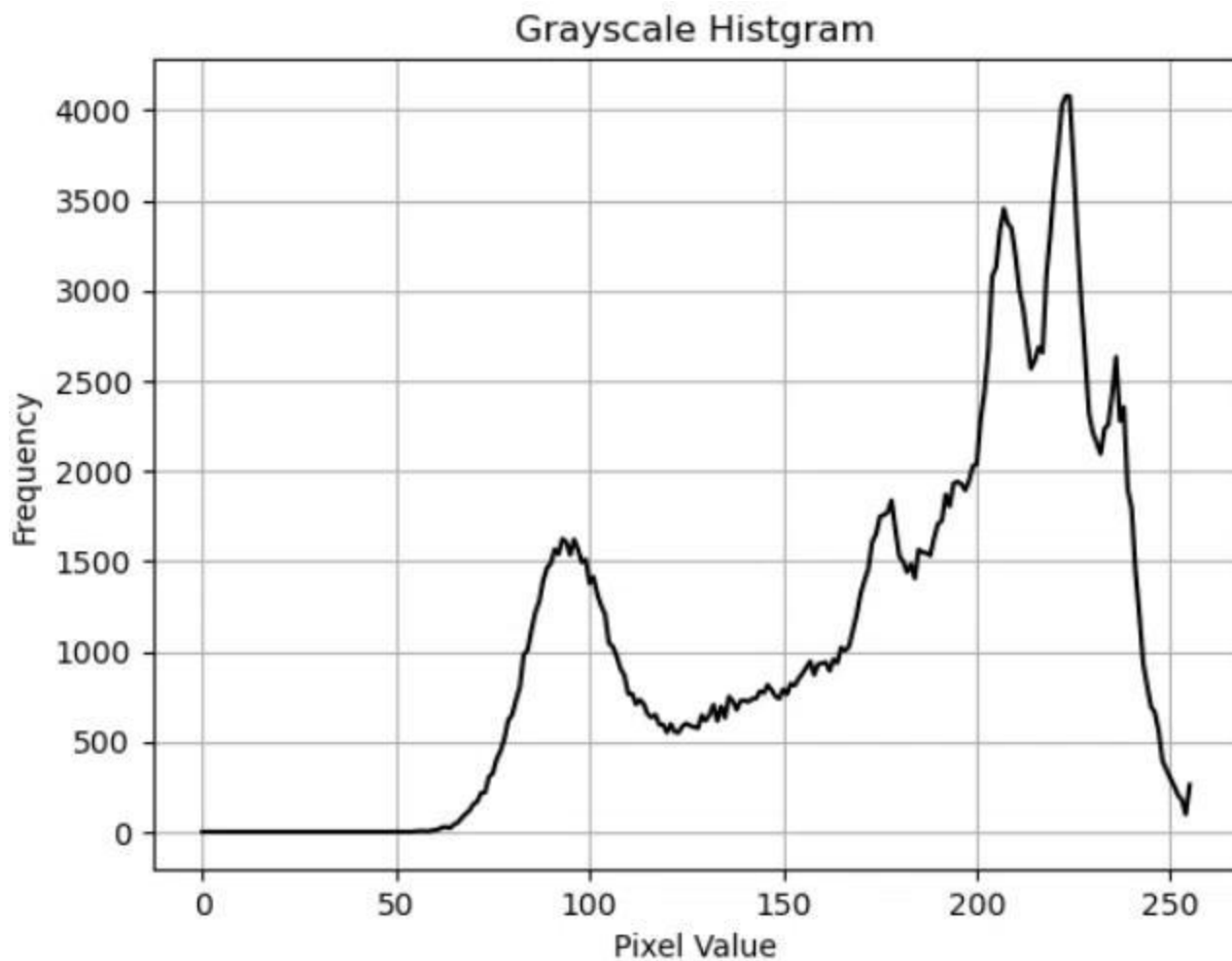
0 Black

128 Gray

255 White

Image Histogram

An image histogram is a graph showing how many pixels belong to each intensity value.

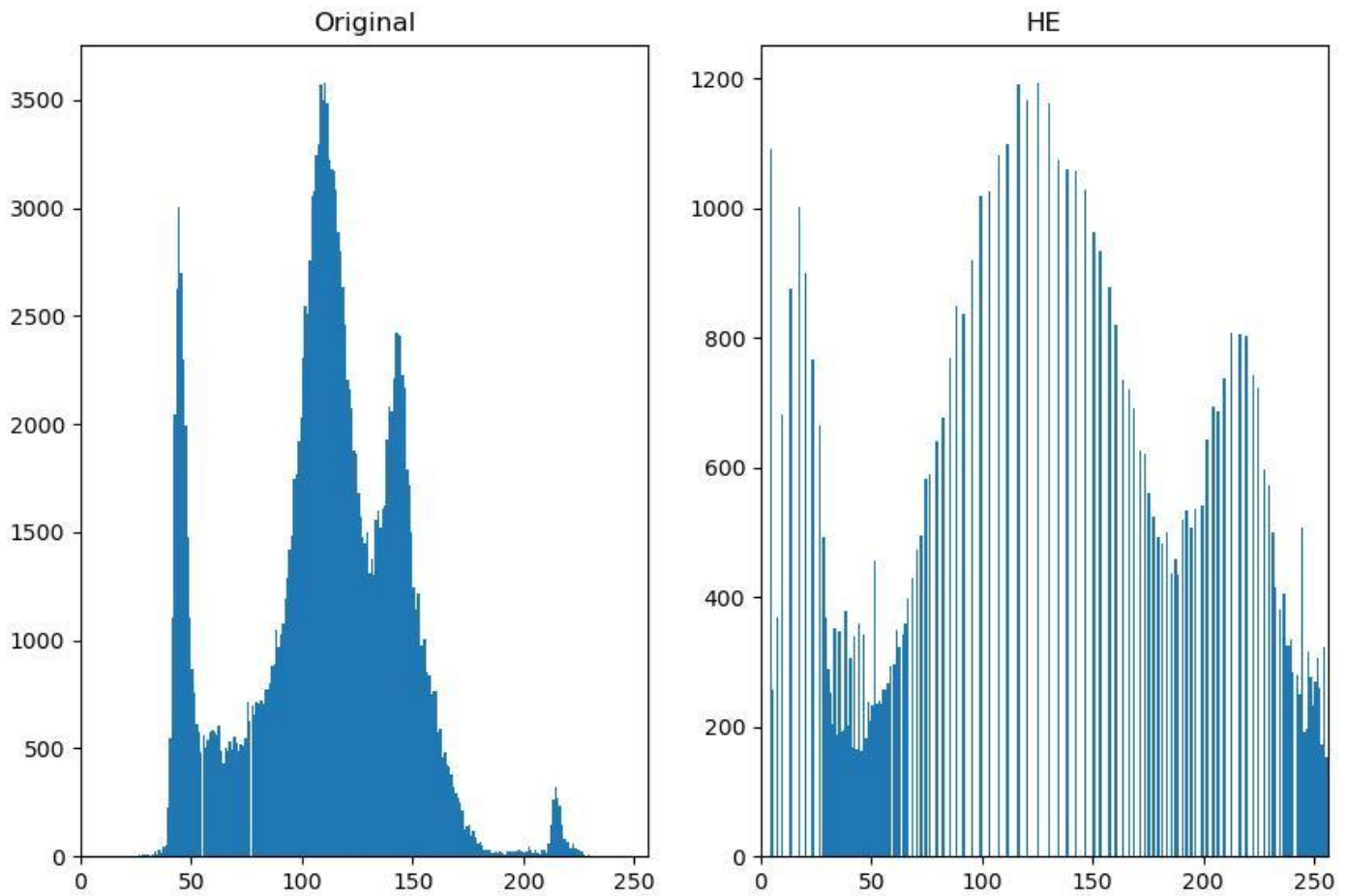


Why Histogram Matters

- Detects low contrast.
- Shows brightness distribution.
- Helps choose enhancement methods.
- Evaluates preprocessing quality.

Histogram Equalization

Histogram Equalization redistributes pixel intensities across the entire range (0–255), improving global contrast.

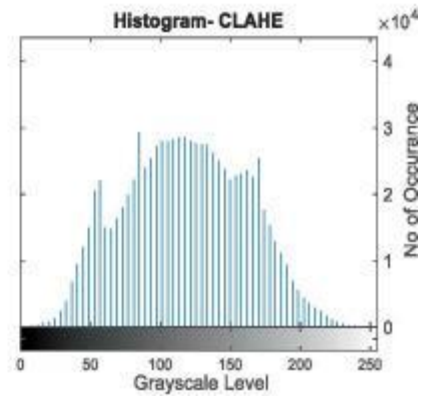
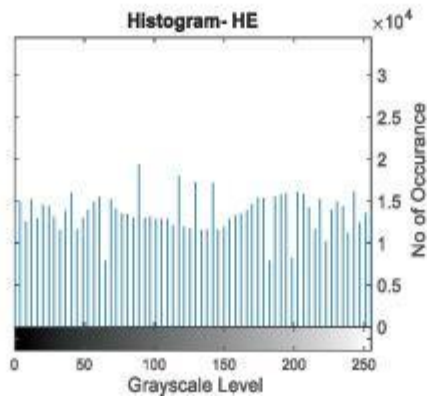
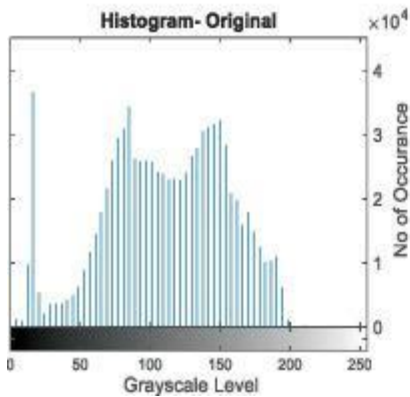


Advantages

- Improves visibility.
- Enhances hidden details.
- Fast preprocessing.
- Suitable for grayscale images.

CLAHE (Contrast Limited Adaptive Histogram Equalization)

CLAHE divides the image into small regions and enhances each region separately while limiting excessive contrast.



Why CLAHE is Better

- Prevents over-enhancement.
- Preserves local details.
- Performs well in uneven lighting.
- Commonly used in medical imaging.

Prerequisites

According to the experiment sheet:

- Basic knowledge of digital image processing
- Python programming
- OpenCV
- NumPy
- Matplotlib
- Understanding of grayscale images
- VS Code / Jupyter Notebook / Google Colab

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Software Required

Software	Purpose
Python 3.x	Programming
OpenCV	Image Processing
NumPy	Numerical Operations
Matplotlib	Plotting Histograms
VS Code/Jupyter Development	

Algorithm

1. Import required libraries.
2. Load a low-contrast image.
3. Convert to grayscale.
4. Display original image.
5. Plot original histogram.
6. Perform contrast stretching.
7. Apply Histogram Equalization.
8. Apply CLAHE.
9. Compare histograms.
10. Analyze results.

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Python Program

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Load image

image = cv2.imread("low_contrast.jpg")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


# Histogram Equalization
```

```
equalized = cv2.equalizeHist(gray)
```

```
# CLAHE
```

```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
```

```
clahe_img = clahe.apply(gray)
```

```
# Contrast Stretching
```

```
stretched = cv2.normalize(gray, None, 0, 255, cv2.NORM_MINMAX)
```

```
# Display Images
```

```
titles = ["Original", "Stretch", "Equalized", "CLAHE"]
```

```
images = [gray, stretched, equalized, clahe_img]
```

```
plt.figure(figsize=(10,8))
```

```
for i in range(4):
```

```
    plt.subplot(2,2,i+1)
```

```
    plt.imshow(images[i], cmap="gray")
```

```
    plt.title(titles[i])
```

```
    plt.axis("off")
```

```
plt.show()
```

```
# Histograms
```

```
plt.figure(figsize=(10,6))
```

```
plt.hist(gray.ravel(),256,[0,256],label="Original")
```

```
plt.hist(equalized.ravel(),256,[0,256],label="Equalized")
```

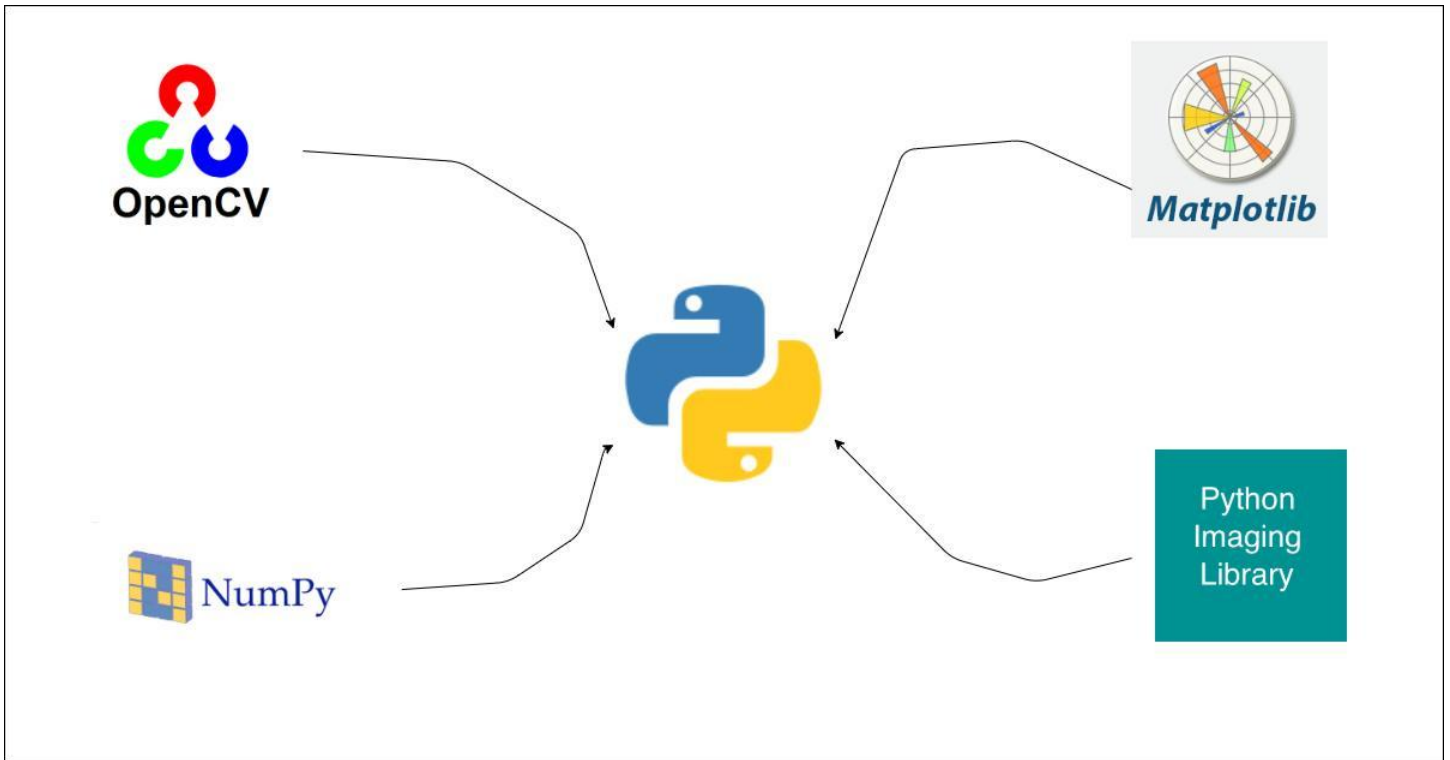
```
plt.hist(clahe_img.ravel(),256,[0,256],label="CLAHE")
```

```
plt.legend()
```

```
plt.show()
```

Step-by-Step Procedure (With Screenshot Placeholders)

Step 1: Import Libraries



Import cv2, NumPy, and Matplotlib.

Screenshot: (Paste your output here)

Step 2: Load Low Contrast Image



Read the input image using `cv2.imread()`.

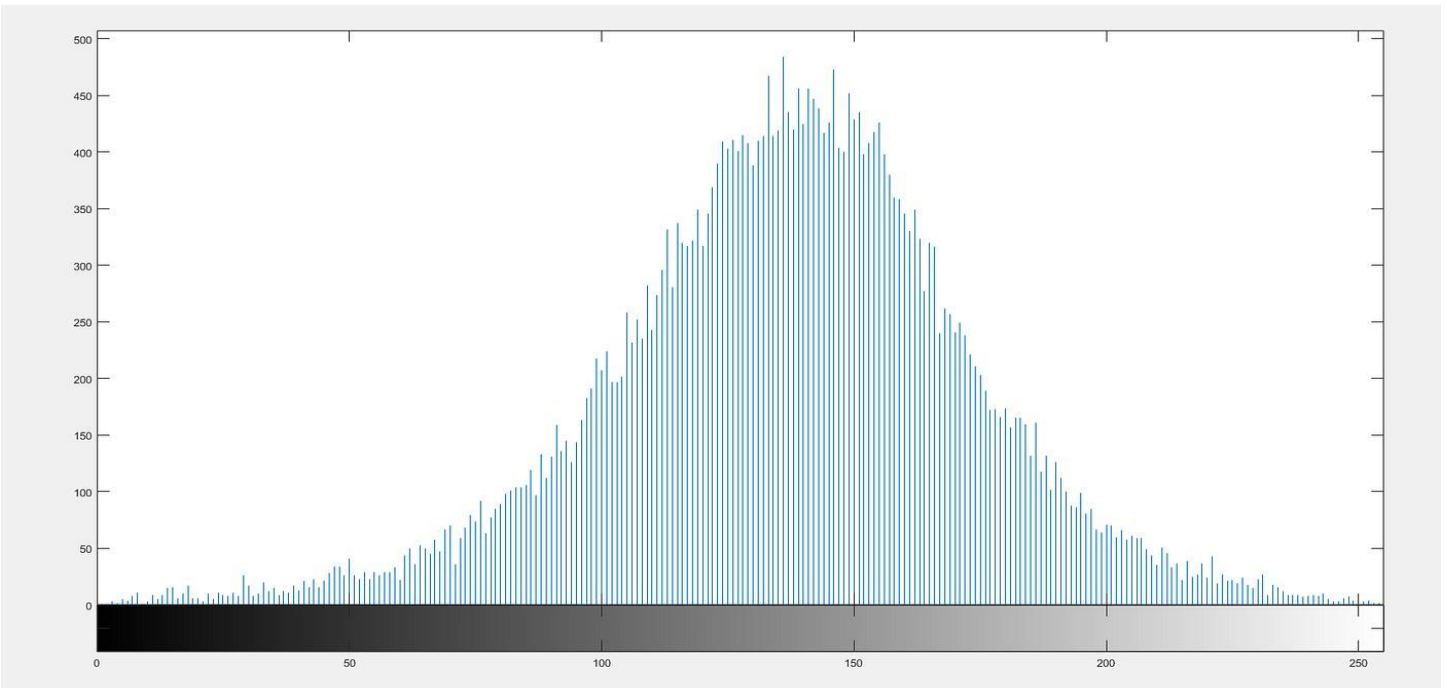
Screenshot: (Paste here)

Step 3: Display Original Image



Show the original grayscale image.

Step 4: Generate Histogram



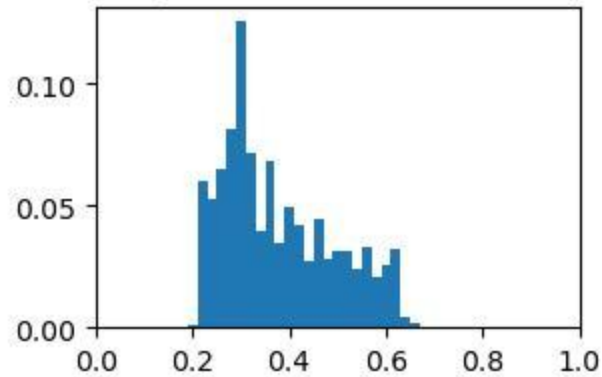
Observe how pixel intensities are distributed.

Step 5: Contrast Stretching

Low contrast original



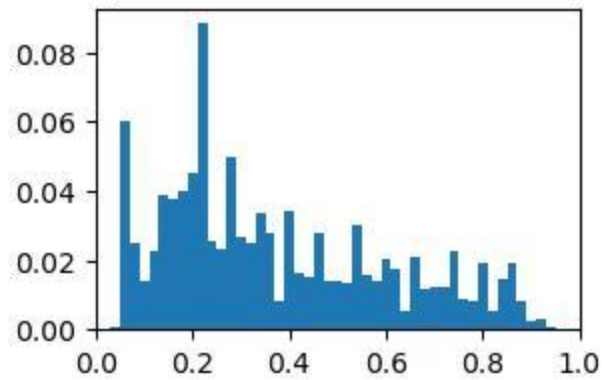
Histogram of low contrast image



Contrast Stretched

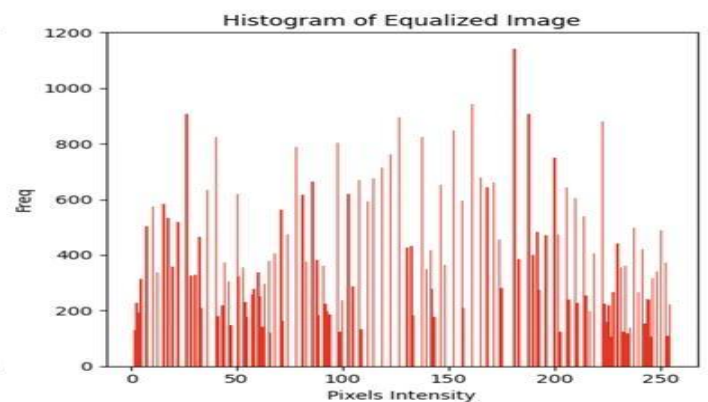
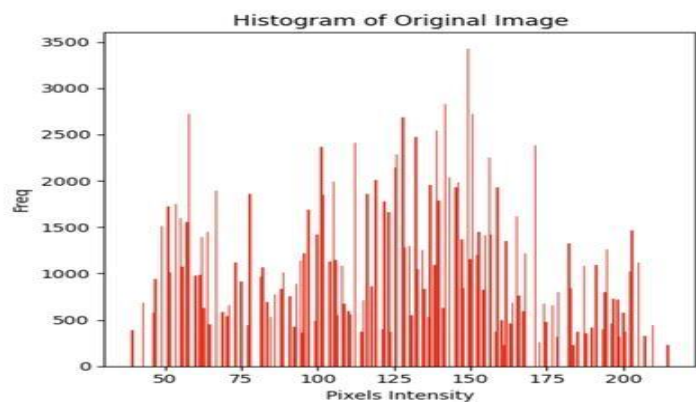


Histogram of contrast stretched image



Improve the dynamic range.

Step 6: Histogram Equalization

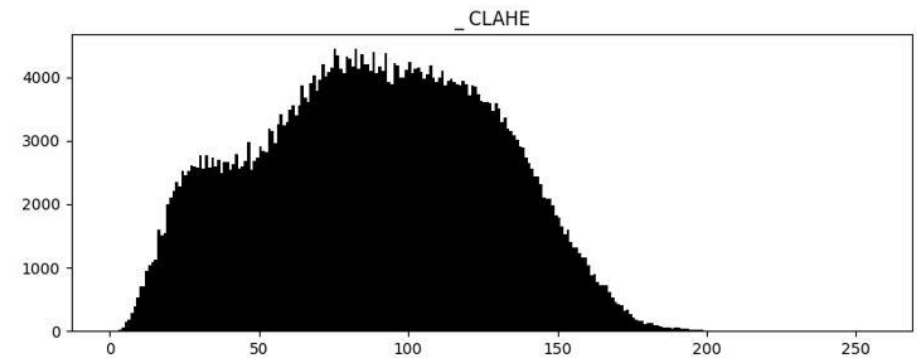
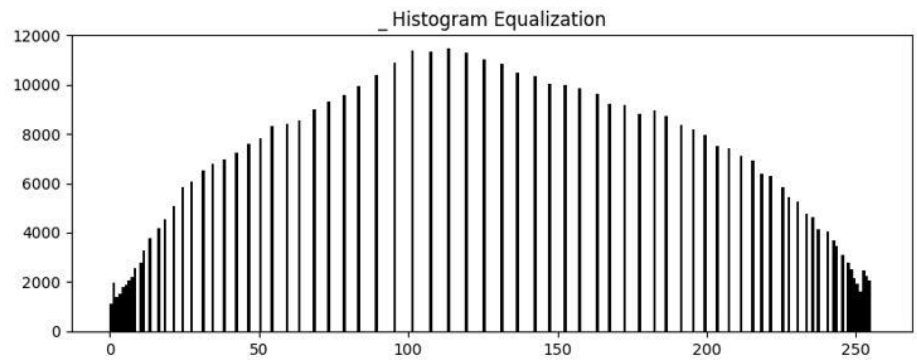
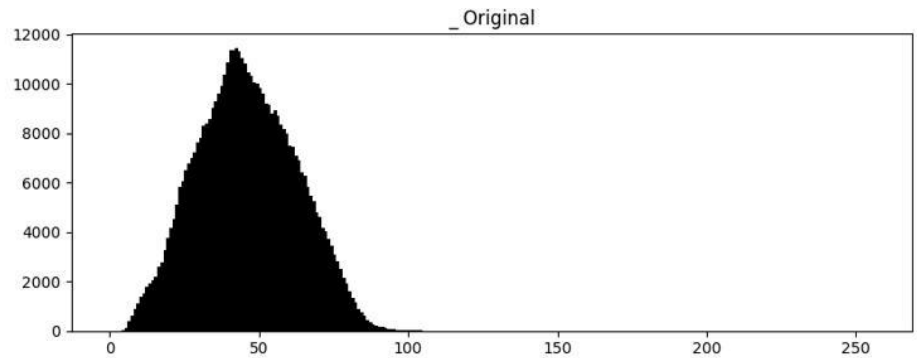


Enhance global contrast.

Step 7: CLAHE

Enhance local contrast.

Step 8: Compare Histograms



Observe the difference.

Observation Table

Operation	Observation	Significance
Original	Low visibility	Needs enhancement
Histogram	Narrow distribution	Low contrast detected
Contrast Stretch	Wider range	Better brightness
Equalization	Improved global contrast	More visible details

CLAHE Better local details Best for uneven lighting

Comparison Table

Feature	Histogram Equalization CLAHE	
Processing	Global	Local
Noise	May increase	Controlled
Detail Preservation	Medium	High
Medical Images	Moderate	Excellent
Uneven Lighting	Poor	Excellent

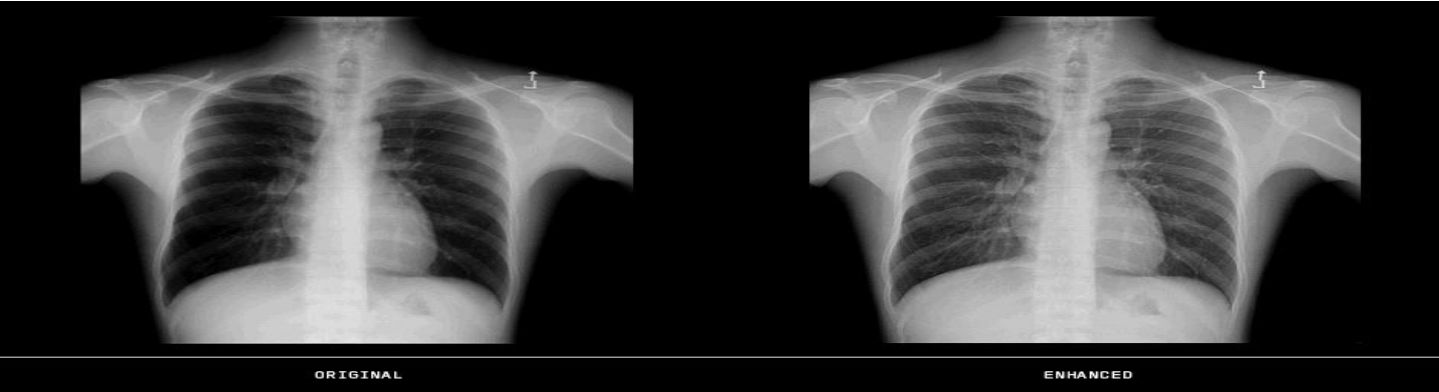
Advantages

- Improves visibility.
- Enhances hidden features.
- Better object detection.
- Faster preprocessing.
- Improves OCR accuracy.
- Supports medical imaging.
- Useful before deep learning.

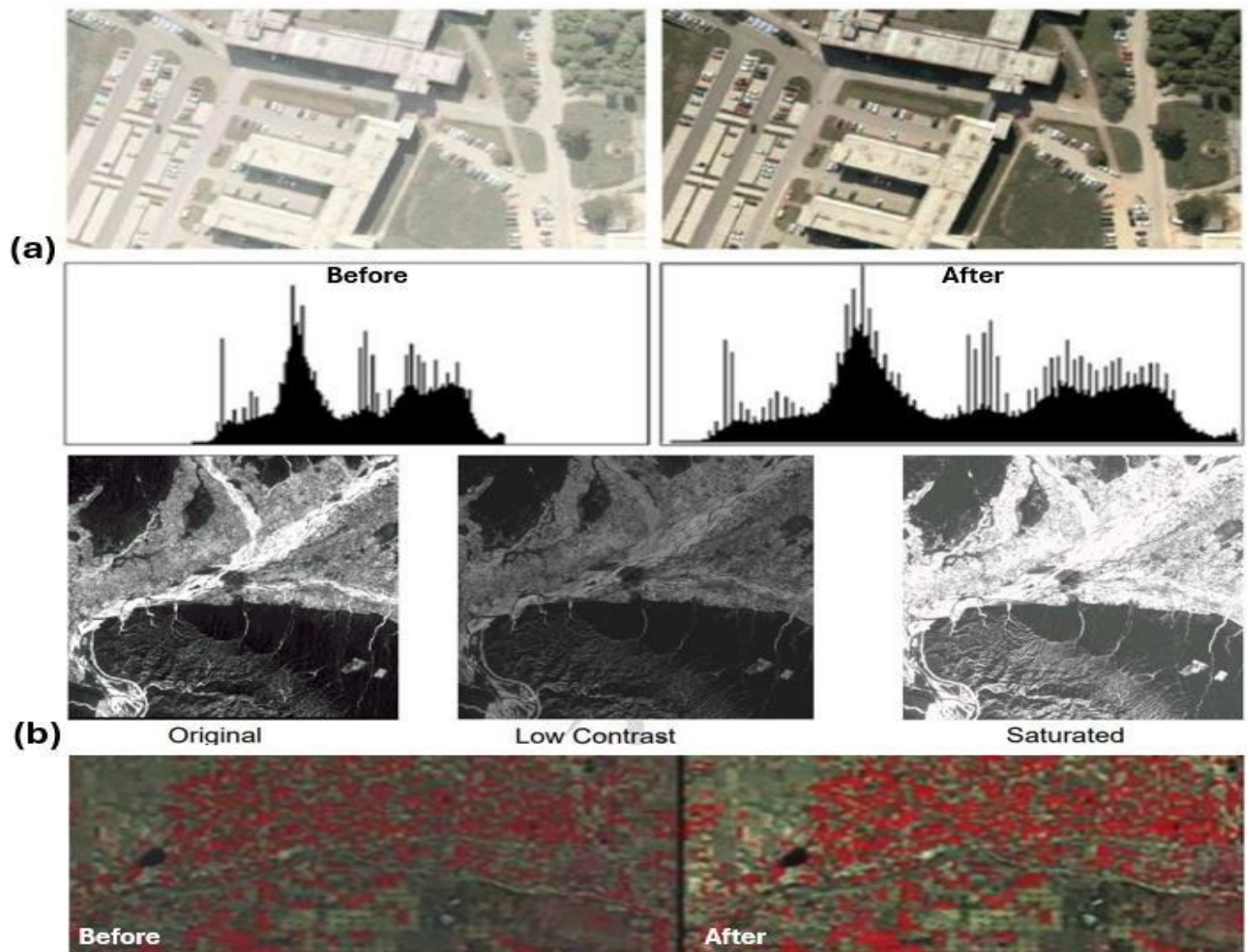
Limitations

- Global equalization may amplify noise.
- Over-enhancement may occur.
- Not suitable for every image.
- Brightness may change.

Real-World Applications



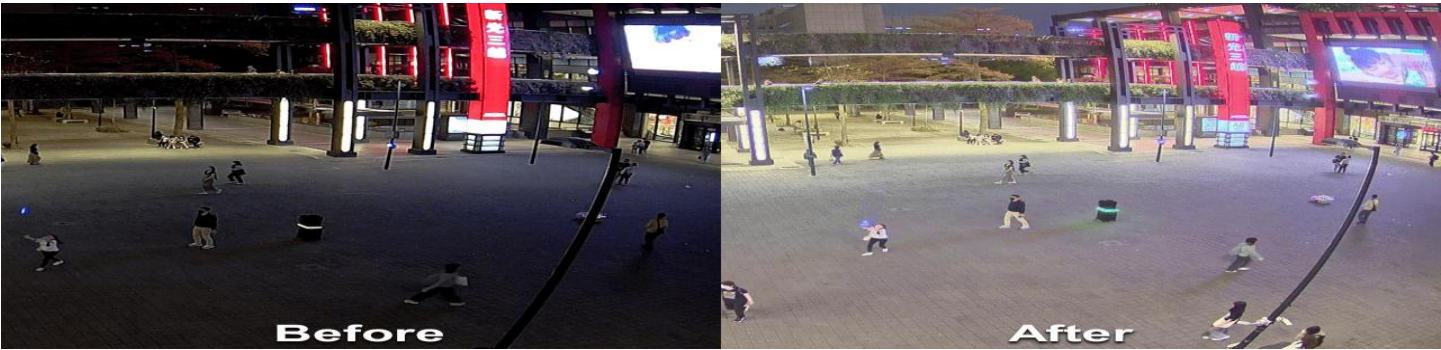
Medical Imaging



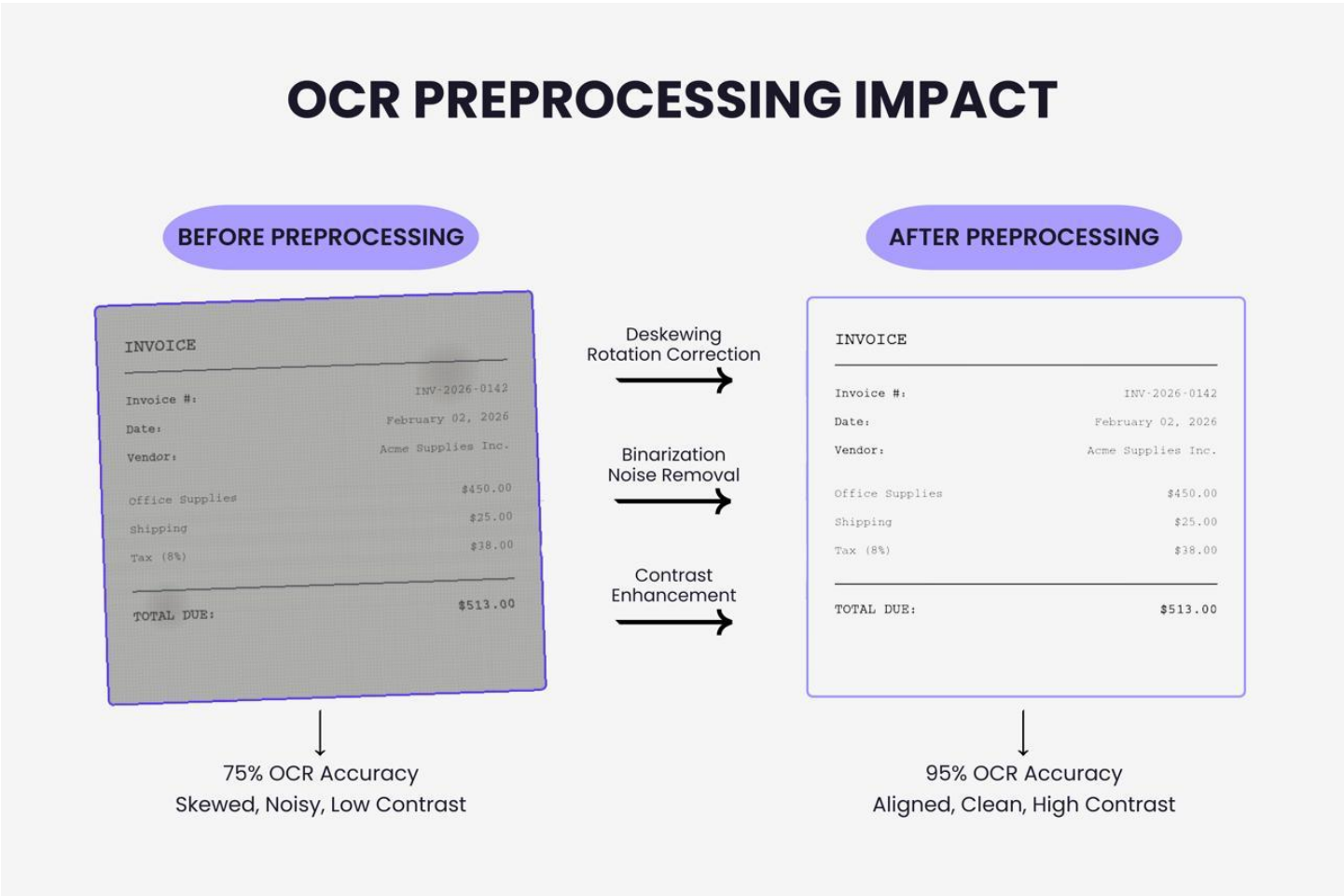
Satellite Images



Autonomous Driving



CCTV Surveillance



OCR & Document Scanning



Industrial Inspection

Other applications include facial recognition, robotics, traffic monitoring, and scientific image analysis.

Precautions

- Use a clear input image.
- Convert to grayscale before equalization.
- Save outputs before closing.
- Use CLAHE carefully for noisy images.
- Compare histograms after every operation.

Result

The experiment was successfully performed using Python and OpenCV. Histogram analysis, Contrast Stretching, Histogram Equalization, and CLAHE were implemented successfully. The results showed that CLAHE produced better local contrast while Histogram Equalization improved overall image visibility, demonstrating the importance of histogram-based preprocessing in computer vision.

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Viva Questions with Answers

1. What is image contrast?

Image contrast is the difference between dark and bright pixels. Good contrast improves visibility and feature extraction.

2. What is an image histogram?

An image histogram shows the number of pixels present at each intensity value.

3. Difference between Contrast Stretching and Histogram Equalization

Contrast Stretching

Expands intensity range

Preserves relative brightness

Histogram Equalization

Redistributes intensities

Improves overall contrast

4. What is CLAHE?

CLAHE is a local contrast enhancement technique that limits excessive amplification of noise.

5. Why apply Histogram Equalization before feature extraction?

It improves edge visibility and increases the accuracy of segmentation and feature detection.

6. Three real-world applications

- Medical imaging
- Satellite image enhancement
- CCTV surveillance

7. Limitations of global Histogram Equalization

It may amplify noise and cannot handle uneven illumination effectively.

8. How does contrast enhancement help object detection?

It makes edges and textures clearer, improving detection accuracy.

9. Histogram-based enhancement vs brightness adjustment

Brightness adjustment changes all pixels equally, while histogram methods intelligently redistribute pixel intensities.

10. Why is CLAHE preferred for medical imaging?

It preserves fine local details while preventing over-enhancement, making it ideal for X-rays, CT scans, and MRI images.